Chapter 1

Swalwell

This appendix records a lightweight physical and historical investigation of the Swalwell corridor segment used throughout the worked example. Its purpose is to provide tangible, route-grounded corroboration for the artefacts produced in Chapter 5, and to act as an evidence trail that partially substitutes for the Step 3.1–3.3 case study within the main method narrative.

The investigation is pragmatic. Within this appendix we triangulate a small number of publicly available historical sources with a brief site visit and opportunistic photography to demonstrate the ODM and registered hazards in the real world in order to support verification of step 3. The evidence used for validation is as follows, figure 1.1, a 1939 corridor map used to localise the scoped segment at Swalwell, showing Swalwell as it was designed originally. Figure 1.2, a surviving photograph of the Swalwell signal box and its surroundings, Figure 1.3, a 1961 document extract indicating closure/operational change and figure 1.4, a photograph taken on a visit to the route, taken from the overpass, directly on route.

https://swalwelluk.co.uk/picpages/pic-signalbox.html https://www.s-r-s.org.uk/html/lnerdiagrams.htm - S707 (c. 1939) http://www.swalwelluk.co.uk/picpages/pic-railmap.html

Figure 1.1 provides the earliest high-resolution, locally specific map used in this thesis to anchor the Swalwell segment. While it is outdated in operational detail, it is valuable for two reasons.

First, it captures the spatial relationship between the junction, adjacent land use, and nearby features in a way that modern signalling schematics do not. Signalling diagrams are authoritative for control structure, but they are not designed to communicate the physical setting that drives certain ecosystem hazards. It helps to explain why this segment is treated as a distinct “micro-domain” within the larger corridor.

Second, the map assists in determining where change influences design, as we progress into Step 6, we understand that the "State of the Art" will need updating regarding information of a given route in order to properly compartmentalize the design. Official documentation

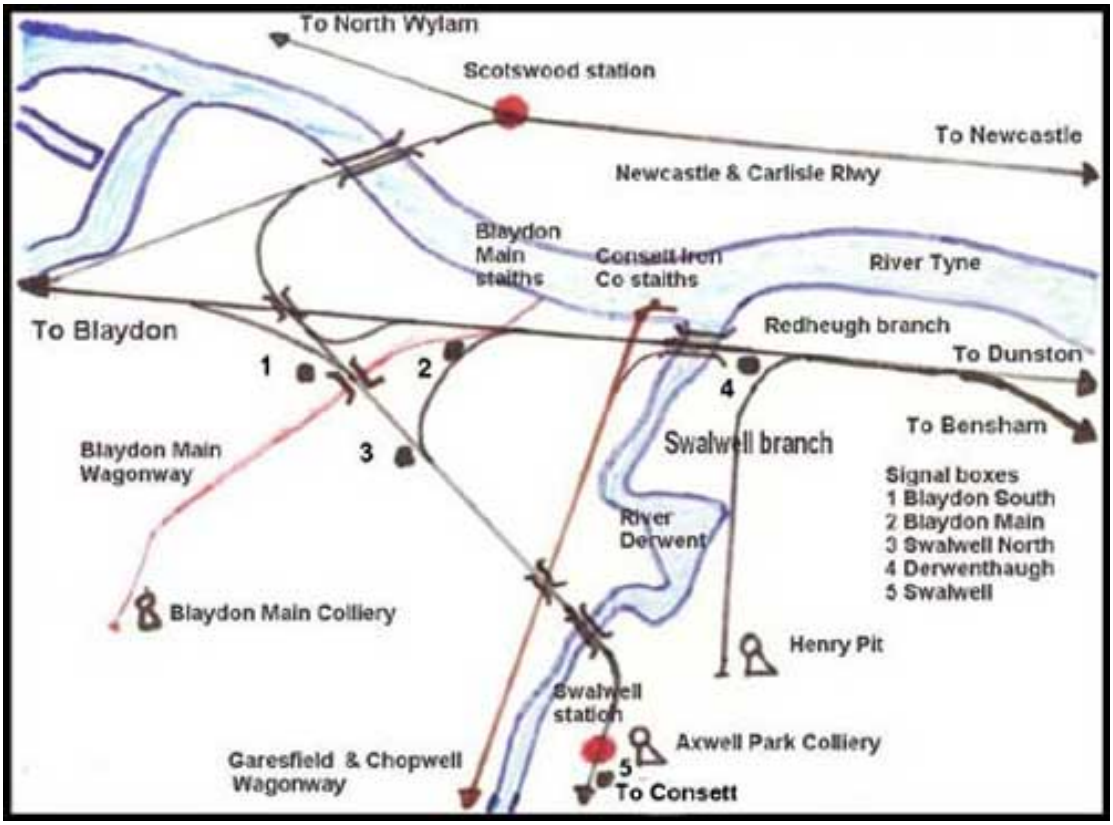


Figure 1.1: Swalwell Junction map - 1939

still refers to out of date or demolished design[1].

Figure 1.2 is the only readily available photograph of the Swalwell signal box, the signal box has closed as of 1985 alongside the surrounding infrastructure and closure of the swalwell station. The image is included due to the relevance of the surrounding of the junction. The image indicates open land use in the immediate vicinity of the railway, consistent with farmland or semi-rural boundary conditions. For the purposes of Step 3, the observation supports the continued inclusion of hazards whose inciting conditions are coupled to adjacent land access and the likelihood of track incursion.

In particular, it provides a route-specific justification for continued inclusion of H06, animal or debris collision. The hazard register frames H06 as an incursion hazard that can manifest during cruise or approach states near vegetation/water or open boundary segments. The signal box image is not proof that animal/debris incursions occur, but it is sufficient to justify that the underlying inciting conditions [2] are present in the scoped domain.

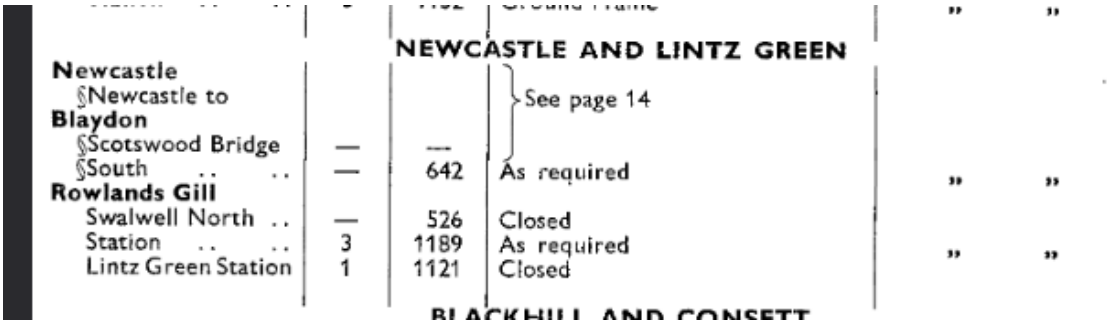
Figure 1.3 shows that the Swalwell branch stop and associated local operational arrangements changed materially between the late 1930s and the early 1960s. [3] However the 'swalwell' label was kept, partly for history, partly because swalwell junction itself was not updated as

[1] see https://www.s-r-s.org.uk/html/lnerdiagrams.htm - S707, the LNER signal box diagram, still referring to Consett railway station, a station which has been out of operation since 1955

[2] open adjacent land, and therefore plausible animal presence and debris sources

[3] Reading contemporary historical documents, it seems most likely that the station closed 1955.

Figure 1.2: Swalwell Signal Box - 1989



			NEWCASTLE AND LINTZ GREEN		
Newcastle					
§Newcastle to			See page 14		
Blaydon					
§Scotswood Bridge	—	—			
§South	—	642	As required	"	"
Rowlands Gill					
Swalwell North ..	—	526	Closed		
Station	3	1189	As required	"	"
Lintz Green Station	1	1121	Closed		

Figure 1.3: Swalwell Junction closed mark - HOURS OF OPENING SIGNAL BOXES 1961

Figure 1.4: Image taken directly above Swalwell Junction overpass, judging distance from overpass to signal

swalwell station was demolished. The junction still exists, however it does not operate as a traditional junction, instead, it switches tracks from an omni-directional track to a single track.

This matters for SACRED because it illustrates a general point that historical artefacts stabilize the current narrative of a route, but are unreliable when regarding the contemporary operation. This is important for Step 6 of the SACRED methodology as it highlights how the state of the art needs to improve in order to properly adjust to legacy technologies[4].

Figures 1.4 and 1.5 show photographs taken from directly above the Swalwell overpass, used here to illustrate the geometric relationship between the overpass and the next signal in the direction of travel. This observation serves two Step 3 purposes.

The presence of an overpass directly above the running line is a route-specific justification for continued inclusion of h10: infrastructure failure. The hazard register treats H10 as a structural integrity hazard that manifests during cruise or approach under a structural element. Secondarily, the same images support H06 again insofar as they reveal nearby vegetation, verges, and the "open" nature of the immediate trackside environment (even if fencing exists elsewhere[5]).

Taken together, these figures show that the Swalwell segment can be locally anchored with corridor-specific artefacts, the environment plausibly supports the inciting conditions for H06 (animal/debris) via open adjacent land use and the environment contains an overpass/bridge feature that justifies H10 as a real, route-specific structural hazard category.

[4]Or it highlights what needs to improve to pave way for the SotA.

[5]Within the UK, dense trees alongside a railway classify as fencing, according to casual correspondence with a railway driver.

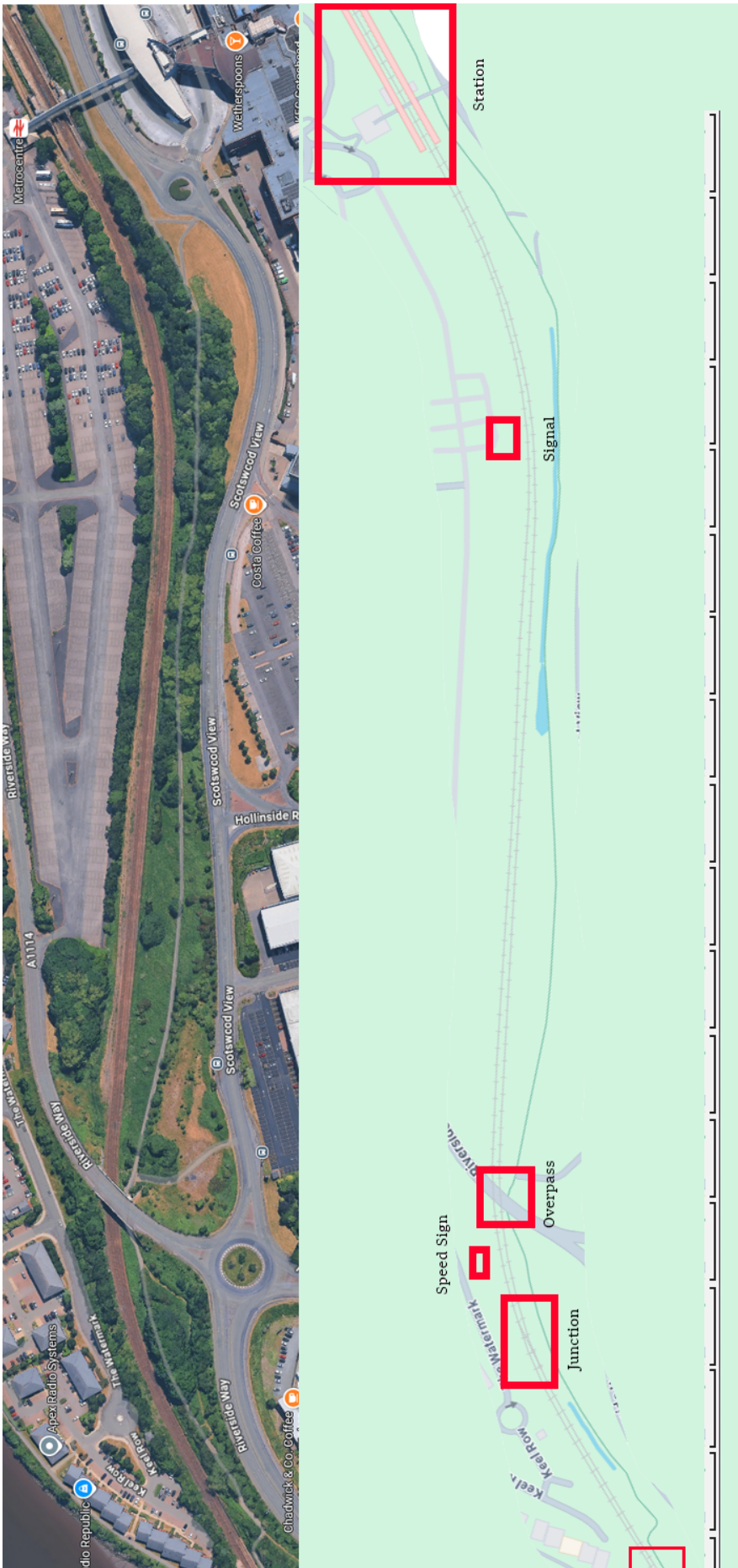
Figure 1.5: Zoomed in version of above.

It is important to note that this appendix does not, and is not intended to, provide physical verification for every hazard listed in Deliverable D6. The hazard register produced in Step 2 deliberately contains a mixture of route-contingent hazards, whose plausibility depends on local geography or infrastructure (e.g., animal incursion, bridge failure) and route-invariant hazards that arise from the fundamental structure of railway operation (e.g., signal passed at danger, authority misinterpretation, or braking misapplication). For the latter category, physical verification at the corridor level is neither necessary nor informative: a SPAD, for example, remains a credible hazard irrespective of route-specific features.

Given the scope of this thesis, Step 3 is applied selectively. Physical and historical investigation is used to corroborate those hazards whose continued inclusion depends on assumptions about the local environment or infrastructure, while route-invariant hazards are retained on the basis of established operational knowledge and prior industry evidence.

http://filestore.limitofshunt.org.uk.s3.amazonaws.com/other-documents/br-ner/signal-boxes-hours-of-opening-1961.pdf

Figure 1.6 shows an annotated version of the google maps overview of the area, matching the ODM, showing the distance estimation between the signal and the speed sign.


Station
Signal
Overpass
Junction
Speed Sign
Metrocentre
Wetherspoons
Costa Coffee
Scotswood View
Hollinside R
Riverside Way
A1114
Apex Radio Systems
Chadwick & Co. Coffee